ALEX FERDICO

MECHANICAL ENGINEERING | AEROSPACE CONCENTRATION | MAY 2027 GRADUATE

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EDUCATION

University at Buffalo, State University of New York — Expected May 2027

Bachelor of Science in Mechanical Engineering | Concentration in Aerospace Engineering

Westchester Community College — May 2024

Associate of Science in Computer Science

TECHNICAL SKILLS

Design & Analysis: SolidWorks, ANSYS, SolidWorks/ANSYS Simulation (FEA), ANSYS Computational Fluid Dynamics (CFD),Fusion 360; CAD assemblies, stress and displacement analysis, mesh refinement

Programming & Tools: Python, MATLAB, C++, Java, OpenCV, MediaPipe, TensorFlow, Arduino

Web Development: HTML, CSS, JavaScript; interactive 3D portfolio development

ENGINEERING LEADERSHIP & EXPERIENCE

Promoting Electric Propulsion (PEP) | E-Board President & Electric Boat Team Lead — 2026–Present

- Lead a student design team developing a remote-controlled electric catamaran for a 2027 competition with a 60 lb payload requirement and a 2-nautical-mile course.
- Develop CAD geometry and perform structural FEA and also CFD of catamaran hull components under payload loading, evaluating stress, displacement, and factor of safety.
- Coordinate propulsion and electrical system design decisions involving brushless motors, electronic speed controllers, battery configurations, power distribution, and disconnect hardware.

SELECTED ENGINEERING PROJECTS

Scissor Jack | Mechanical Design & Structural FEA — SolidWorks

- Modeled scissor jack components and assembly interfaces; assessed part geometry, load paths, and mechanical fit.
- Applied loads and boundary conditions in structural simulation to study stress, deformation, mesh sensitivity, and potential component redesign.

Computer Vision Hand-Gesture Recognition — Python | OpenCV | MediaPipe | TensorFlow

- Developed a computer vision project using machine-learning tools to recognize hand gestures and track hand landmarks in three dimensions.
- Integrated coordinate tracking and confidence-based prediction output to support real-time gesture interpretation.

Interactive Engineering Portfolio — HTML | CSS | JavaScript | Spline

- Designed and coded alexferdico.com to present technical projects, incorporating interactive 3D graphics and custom website navigation.

ADDITIONAL WORK EXPERIENCE

The Garrison | Food Runner / Waiter | Garrison, NY — Summer 2023

Beekman Golf Course | Groundskeeper | East Fishkill, NY — Summer 2022